Time-Complexity of Key-Insertion in Python Dicts

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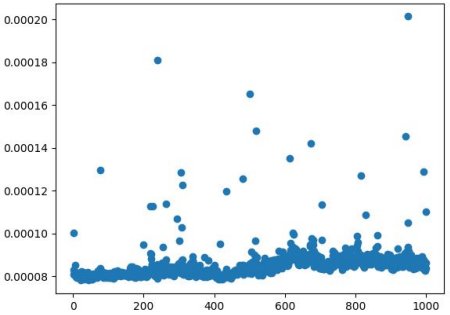
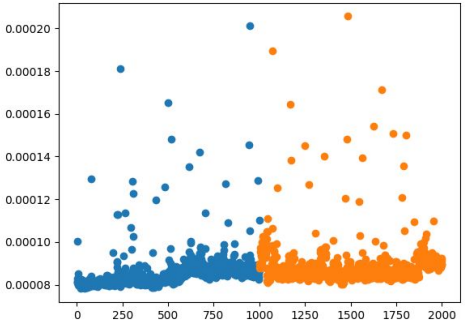
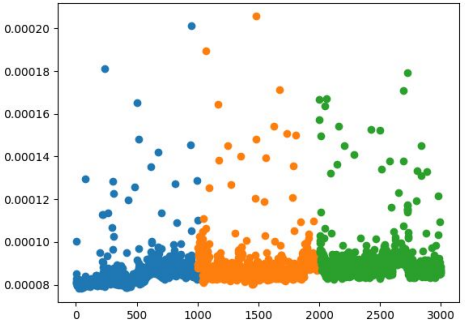
Initial bucket sizes showed O(1)

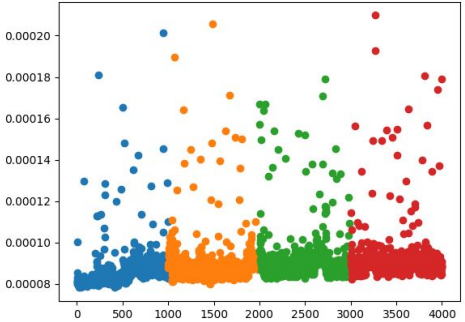
```
(0,    1000)
(1000, 2000)
(2000, 3000)
(3000, 4000)
(4000, 5000)
```

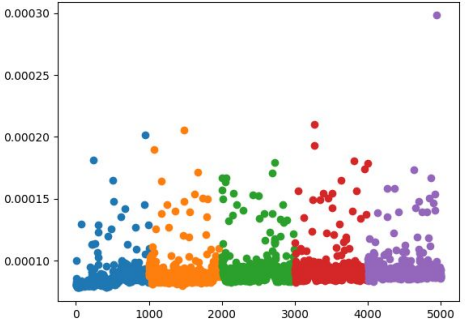
Too similar











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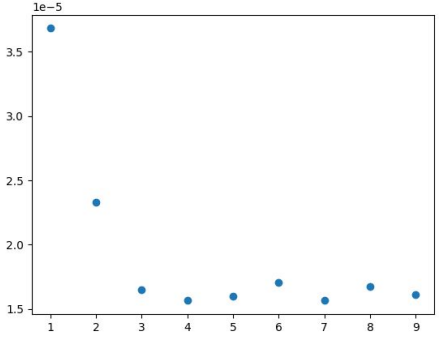
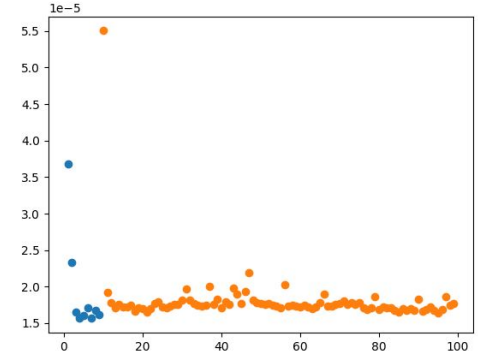
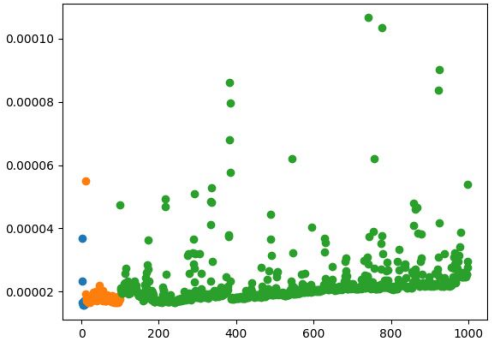
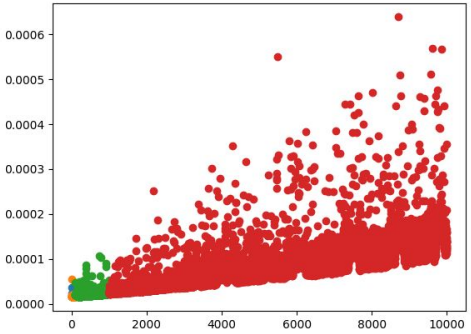
Exponentially larger buckets indicate O(N)

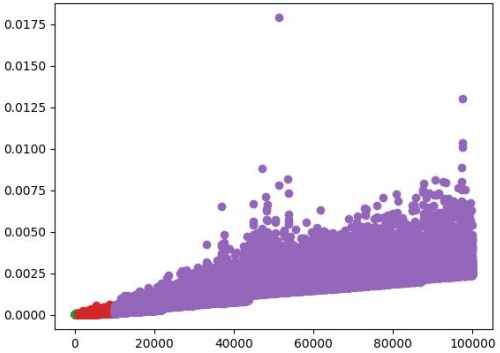
```
(1, 10)
(10, 100)
(100, 1000)
(1000, 10000)
(10000, 100000)
```











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Data summary from five bucket runs

```
data = [
    {'size_range': [1, 10], 'mean_time': np.float64(1.9334217843910057e-05), 'std_time': np.float64(6.577901417405355e-06)},
    {'size_range': [10, 100], 'mean_time': np.float64(1.8043926037434075e-05), 'std_time': np.float64(4.022633755827987e-06)},
    {'size_range': [100, 1000], 'mean_time': np.float64(2.2379783654792442e-05), 'std_time': np.float64(8.042521196087182e-06)},
    {'size_range': [1000, 10000], 'mean_time': np.float64(8.860559031988184e-05), 'std_time': np.float64(5.263201767485302e-05)},
    {'size_range': [10000, 100000], 'mean_time': np.float64(0.0015507531675199668), 'std_time': np.float64(0.0008694342600839186)},
]

res = profileTcoki(data)
```

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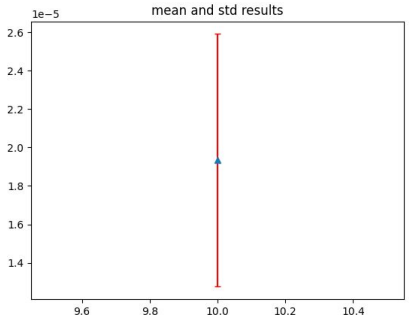
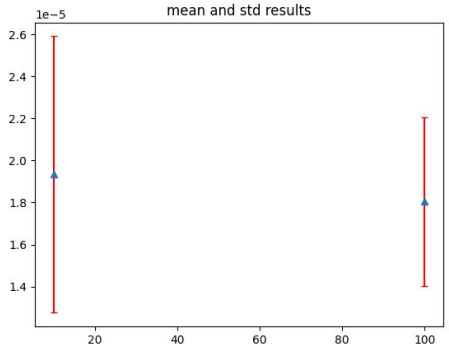
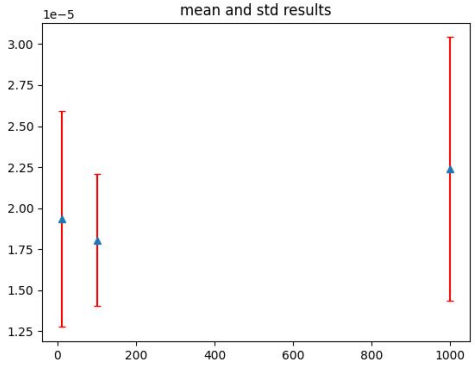
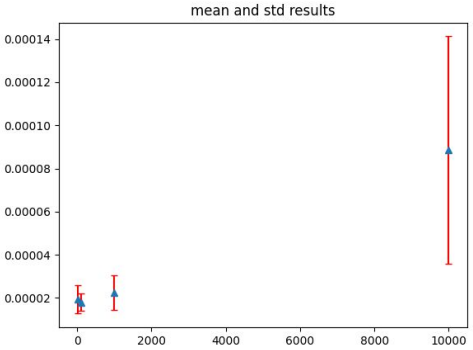
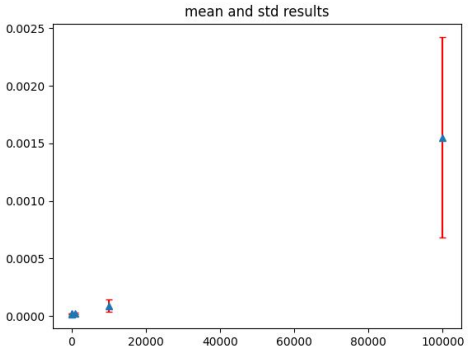
Plots of mean and standard deviation per run











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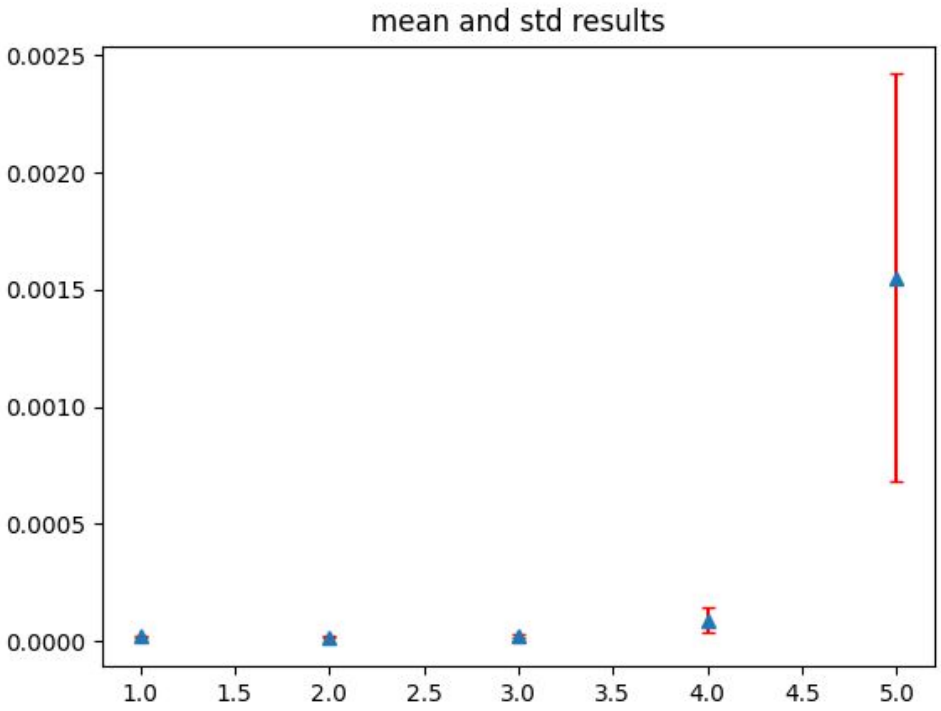
Mean and standard deviation summarized

```
(1, 10)
(10, 100)
(100, 1000)
(1000, 10000)
(10000, 100000)
```



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The time-complexity of key-insertion in Python dictionaries is constant complexity.

$$O(1)$$

However, as the dictionary size gets very large the time-complexity approaches linear complexity.

$$O(N)$$

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